Problem A: Touko's Perfection

Time Limit: 2.0s | Memory Limit: 256MB

Description

"I have to be perfect. Everyone expects it."

Touko Nanami is obsessing over the Student Council's workflow. She has quantified the "Competence Level" of the N members of the extended council (including volunteers) as a sequence $A_1, A_2, \ldots, A_N$.

For the workflow to be seamless, the competence levels of the members must be **non-decreasing**. That is, the competence of the first member must be less than or equal to the second, and so on ($B_1 \leq B_2 \leq \cdots \leq B_N$).

Touko can "train" any member to change their competence level from A_i to any integer B_i. The effort required to change a member's competence is $|A_i - B_i|$.

Touko wants to minimize the **total effort** $\sum_{i=1}^{N} |A_i - B_i|$ required to make the sequence non-decreasing.

Input

The first line contains a single integer N ($1 \leq N \leq 2 \cdot 10^5$). The second line contains N integers $A_1, A_2, \ldots, A_N$ ($1 \leq A_i \leq 10^9$).

Output

Output the minimum total effort required.

Examples

Example 1

Input:

```
5
3 2 5 1 7
```

Output:

```
5
```

Explanation: We can modify the sequence to $2, 2, 5, 5, 7$. Cost: $|3-2|+|2-2|+|5-5|+|1-5|+|7-7| = 1+0+0+4+0 = 5$. Alternatively, $2, 2, 2, 5, 7$ costs $|3-2|+|2-2|+|5-2|+|1-5|+|7-7| = 1+0+3+4+0 = 8$, which is worse.

Example 2

Input:

```
4
1 1 1 1
```

Output:

```
0
```